

Inventory Dashboard Query Checklist

Printable workflow tracker for report/query development

How to use this checklist: Work from top to bottom. Mark a query as done only after it runs without errors, returns sensible test output, and you know which report or dashboard screen it belongs to.

Current design rule: Keep Weeks on Hand, Inventory Exceptions, Financial Performance, and Order Scenario queries separate. Combine them only when a specific dashboard screen needs them together.

1. Query foundation and sanity checks

Done	Query / Function to Build	Purpose	Notes / Acceptance Check
☐	create_connection()	Open the SQLite database and enable foreign keys.	Connection works; no error; PRAGMA foreign_keys = ON.
☐	show_tables()	Confirm the schema exists after running create_database.py.	Tables display, including Product, InventorySnapshot, SalesHistory, ProductFinancial, and ImportBatch.
☐	show_import_batch_count()	Verify import batches are being created.	Count increases after inventory and sales imports.
☐	get_latest_import_batch_id(import_type)	Reusable helper to find the newest import batch by type.	Returns latest Inventory or SalesHistory ImportBatchId.
☐	show_product_count()	Confirm products were imported from the inventory file.	Expected count matches cleaned inventory rows.

2. Import quality and data validation queries

Done	Query / Function to Build	Purpose	Notes / Acceptance Check
☐	show_duplicate_skus()	Find duplicate SKUs in source data or database.	Should return none after cleaning invalid XX rows.
☐	show_missing_sales_products()	Find SKUs in SalesHistory import that do not exist in Product.	Useful for missing POS items or sales report oddities.
☐	show_products_without_sku()	List planned/manual products that do not have POS SKUs yet.	Should include potential new items entered during order planning.
☐	show_products_not_in_pos()	List products where IsInPOS = 0.	Used for planned items not yet confirmed in POS.
☐	show_products_in_pos()	Verify imported POS products are marked IsInPOS = 1.	Inventory import should set IsInPOS to 1.
☐	show_latest_inventory_batch_summary()	Count records in latest inventory batch.	Count equals latest InventorySnapshot rows for latest Inventory ImportBatchId.
☐	show_latest_sales_batch_summary()	Count records in latest sales batch.	Count equals latest SalesHistory rows for latest SalesHistory ImportBatchId.

3. Inventory and stock health reports

Done	Query / Function to Build	Purpose	Notes / Acceptance Check
☐	show_latest_inventory_snapshot()	Display current inventory based on latest inventory import.	Uses latest Inventory ImportBatchId only; starts with LIMIT 10 or 20.
☐	show_weeks_on_hand_report()	Main operational WOH report for active stocked items.	Focus on QtyOnHand, UnitsSold, and EstimatedWeeksOnHand. Keep financials separate.

Done	Query / Function to Build	Purpose	Notes / Acceptance Check
[]	show_low_weeks_on_hand()	Rank items most likely to need ordering soon.	Filter UnitsSold > 0; sort EstimatedWeeksOnHand ascending.
[]	show_inventory_exceptions()	Find negative or suspicious on-hand quantities.	Includes QtyOnHand < 0 and items sold while inventory is zero or negative.
[]	show_zero_or_negative_stock_with_sales()	Catch items that sold even though stock was <= 0.	Useful for missed swaps, wrong SKU sales, or inventory adjustment issues.
[]	show_no_recent_sales()	Find items with no sales over the chosen period.	Later use 6-month lookback so old/seasonal items do not clutter WOH.
[]	show_active_stocked_skus()	Approximate the 325-ish SKUs currently stocked.	Start with QtyOnHand > 0 OR recent UnitsSold > 0; refine later with IsActive/IsRegularStock if added.

4. Sales history reports

Done	Query / Function to Build	Purpose	Notes / Acceptance Check
[]	show_latest_sales_history()	Display latest sales records joined to product names.	Uses latest SalesHistory ImportBatchId.
[]	show_returns_report()	Show products with UnitsSold < 0.	Useful for returns, discounts, or correction spot checks.
[]	show_top_units_sold()	Rank products by quantity sold.	Can be latest week first, then later by date range.
[]	show_sales_by_product_date_range()	Summarize sales for a product across multiple sales periods.	Needed before reliable 4-week, 8-week, or 6-month averages.
[]	show_sales_velocity()	Calculate average weekly units sold.	Later basis for better Weeks on Hand: QtyOnHand / AvgWeeklyUnitsSold.

5. Financial performance reports

Done	Query / Function to Build	Purpose	Notes / Acceptance Check
[]	show_latest_product_financials()	Display latest cost, retail, margin percent, and margin dollars.	Uses latest ProductFinancial rows from latest inventory import.
[]	show_top_performers_by_profit()	Rank products by estimated gross profit.	UnitsSold * MarginDollars; keep separate from WOH report.
[]	show_top_performers_by_units()	Rank high-volume sellers with financial context.	Useful for comparing volume vs profitability.
[]	show_low_margin_items()	Find items with weak margin percent or margin dollars.	Helps review pricing or supplier cost changes.
[]	show_price_or_cost_changes()	Compare latest financial import against prior import.	Later report; useful when vendor costs or retail prices change.

6. Order scenario and weekly order planning queries

Done	Query / Function to Build	Purpose	Notes / Acceptance Check
[]	show_order_scenarios()	List saved order scenarios.	ScenarioName, date, vendor, sales rep, notes, and budget if added.
[]	show_order_scenario_detail(scenario_id)	Display all products planned in one scenario.	Includes cases/bottles, cost estimate, comments, and delivery date if fields are added.
[]	show_order_scenario_totals(scenario_id)	Calculate total cases, bottles, and estimated dollars.	Used to keep weekly order within budget.

Done	Query / Function to Build	Purpose	Notes / Acceptance Check
[]	show_budget_remaining(scenario_id)	Compare estimated order total against budget.	BudgetAmount - TotalEstimatedCost.
[]	show_projected_woh_after_order(scenario_id)	Estimate WOH after adding planned order quantities.	Does not change real inventory; planning only.
[]	show_planned_products_not_in_pos()	Show new items in scenarios that are not yet in the POS.	IsInPOS = 0 and SKU is NULL.
[]	show_incoming_inventory_by_delivery_date()	Help plan space for what is coming into the store.	Groups scenario items by delivery date/vendor.

7. Maintenance and admin queries

Done	Query / Function to Build	Purpose	Notes / Acceptance Check
[]	search_product_by_sku_or_name()	Find products quickly for maintenance screens.	Search by SKU, partial product name, or planned item name.
[]	show_products_missing_category()	Find products that still need category or subcategory assignment.	SubCategoryId IS NULL.
[]	show_products_missing_vendor_or_rep()	Find products missing vendor or sales rep assignment.	VendorId IS NULL OR SalesRepId IS NULL.
[]	show_seasonal_products()	List products marked IsSeasonal = 1.	Useful for products that may return later.
[]	show_discontinued_products()	List products marked IsDiscontinued = 1.	Useful to exclude from normal order planning.
[]	match_planned_product_to_pos_sku()	Update a planned product once IT assigns a SKU.	Sets SKU and IsInPOS = 1 on the existing planned Product row.

8. Dashboard card / KPI queries

Done	Query / Function to Build	Purpose	Notes / Acceptance Check
[]	get_total_active_skus()	Count currently active/stocked SKUs.	Start with QtyOnHand > 0 OR recent sales > 0.
[]	get_low_woh_count()	Count items below a chosen WOH threshold.	Example: EstimatedWeeksOnHand < 2.
[]	get_inventory_exception_count()	Count items needing inventory review.	Negative QtyOnHand, zero stock with sales, or other exception logic.
[]	get_latest_sales_total_units()	Total units sold in latest sales batch.	Quick weekly sales KPI.
[]	get_order_scenario_total_budget()	Total estimated dollars in a selected order scenario.	Used for order planning dashboard card.

9. Testing and cleanup checklist for each query

Done	Query / Function to Build	Purpose	Notes / Acceptance Check
[]	Start with LIMIT 10 or LIMIT 20	Keep early output readable while testing.	Remove or adjust limit only after query is validated.
[]	Print clear headers	Make terminal output easy to read.	Use formatted strings or a helper function.
[]	Test with known SKU examples	Verify the query matches real expected products.	Pick one product you can manually verify.
[]	Handle NULL/zero/negative cases intentionally	Avoid hiding useful exceptions by accident.	Decide whether each report should include or exclude them.
[]	Commit clean checkpoints	Save progress when a query works.	Commit after grouped, tested changes.
[]	Refactor shared helpers later	Avoid copying repeated code forever.	Move common DB/report helpers into shared modules after the logic works.